

Paddy Irrigation Expert Questionnaire

This questionnaire collects expert knowledge about paddy cultivation water requirements. Your answers will help create a smart irrigation system for paddy farmers.

Estimated time: 20-30 minutes.

Thank you for your valuable input!

* Indicates required question

Expert Background Information

1. **Name:** *

2. **Current Position/Role:** *

3. **Years of Experience in Paddy Cultivation:** *

4. **Primary Location/Region Where You Work:** *

5. **Type of Expertise:** (Check all that apply) *

Check all that apply.

- ☐ Practical farming/field management
- ☐ Agricultural research
- ☐ Extension services/farmer training
- ☐ Irrigation management
- ☐ Other: _____

6. **Approximate Number of Paddy Farms/Projects You Have Worked With:** *

Section 1: Soil Moisture Requirements

Questions about soil moisture levels for different crop stages:

7. For the **seedling stage** (first 10-20 days after planting), what is the ideal soil moisture level? What is the minimum and maximum acceptable level? *

8. For the **tillering stage** (when rice plants start forming side shoots), what is the ideal soil moisture level? What are the minimum and maximum acceptable levels? *

9. For the **vegetative growth stage** (active leaf and stem growth), what is the ideal soil moisture level? What are the minimum and maximum acceptable levels? *

10. For the **reproductive stage** (flowering and grain formation), what is the ideal soil moisture level? What are the minimum and maximum acceptable levels? *

11. For the **ripening stage** (grain maturation), what is the ideal soil moisture level? What are the minimum and maximum acceptable levels? *

12. At what soil moisture level should we consider the crop to be under water stress (too dry)? *

13. At what soil moisture level is there too much water that could harm the crop? *

Section 2: Crop Growth Stages

Questions to understand crop development:

14. How many days typically does each crop stage last in paddy cultivation? *

Seedling stage (in days)

15. Tillering stage (in days) *

16. Vegetative stage (in days) *

17. Reproductive stage (in days) *

18. Ripening stage (in days) *

Section 3: Water Requirements and Tank Levels

Questions about water availability and usage:

19. What is the minimum water level required in the tank to safely irrigate the paddy field? *

20. How much water (in liters or cubic meters) is typically needed per square meter of paddy field for each crop stage? *

21. If the tank water level is low, should irrigation be reduced or stopped at certain crop stages? Which stages are most critical?

Section 4: Rainfall Impact

Questions about rainfall and irrigation decisions:

22. For how long in advance should we predict rainfall to make irrigation decisions? *
(For example: next 6 hours, next 12 hours, next 24 hours, or next 48 hours)

23. If rainfall is predicted for the duration mentioned above, at what rainfall amount (in *
millimeters) should we reduce or skip irrigation?

24. Can too much rainfall harm the crop at certain stages? Which stages are most
sensitive?

Section 5: Evapotranspiration (Water Loss)

Questions about water loss from plants and soil:

25. On hot, sunny days (high temperature), does the crop need more water than *
usual? How much more (as a percentage)?

26. On cooler, cloudy days, does the crop need less water? How much less (as a percentage)? *

27. During which crop stage does the plant lose the most water through its leaves? *

28. What evapotranspiration range(in mm) is considered "normal" for paddy cultivation in your region? What is "hot" and what is "cold"? *

Section 6: Decision Rules for Irrigation

Questions about when and how much to irrigate:

29. If soil moisture is below the ideal level, but rain is expected tomorrow, should we still irrigate today? What factors help make this decision?

30. If the tank water level is low and soil moisture is decreasing, which crop stage should receive priority for irrigation?

31. If soil moisture is adequate but evapotranspiration is high (hot day), should we add extra water as a precaution?

32. What is the maximum amount of water that should be added in a single irrigation event to avoid flooding?

33. Should irrigation be done at a specific time of day? Does morning vs. evening irrigation make a difference?

Section 7: Local Practices and Experience

Questions about practical knowledge:

34. Based on your experience, what are the most common mistakes farmers make regarding irrigation timing?

35. What early warning signs indicate that a paddy field is getting too dry or too wet?

36. In your region, how often (how many times per week) is irrigation typically needed during each crop stage under normal weather conditions?

37. Are there traditional methods or rules of thumb that experienced farmers use to decide when to irrigate?

Section 8: Output Expectations

Questions about the CWR prediction:

38. In your expert opinion, how would you rate the potential effectiveness of this system for paddy cultivation? (Please rate on a scale of 1-5, where 1 = Not effective, 5 = Very effective)

Mark only one oval.

1	2	3	4	5
☐	☐	☐	☐	☐

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